

PROJECT ONE - Analysis of Student Success Data

BACKGROUND AND OVERVIEW

This project investigates the relationship between psychological stressors and academic outcomes among university students. Using SQL to merge and analyze disparate datasets, the study aims to identify at-risk student populations and evaluate the effectiveness of institutional support systems.

DATA STRUCTURE AND OVERVIEW

The analysis was performed in Google BigQuery using a relational schema consisting of two primary datasets:

student_performance_finalscore		student_performance	
Student_ID	STRING	Student_ID	STRING
Age	INT64	Age	INT64
Gender	STRING	Gender	STRING
Hours_Studied	NUMERIC	Hours_Studied	NUMERIC
Attendance	NUMERIC	Attendance	NUMERIC
Sleep_Hours	NUMERIC	Sleep_Hours	NUMERIC
Stress_Level	NUMERIC	Stress_Level	NUMERIC
Screen_Time	NUMERIC	Screen_Time	NUMERIC
Previous_GPA	NUMERIC	Previous_GPA	NUMERIC
Part_Time_Job	STRING	Part_Time_Job	STRING
Study_Method	STRING	Study_Method	STRING
Diet_Quality	STRING	Diet_Quality	STRING
Internet_Quality	STRING	Internet_Quality	STRING
Extracurricular	STRING	Extracurricular	STRING
Tutoring_Sessions_Per_Week	INT64	Tutoring_Sessions_Per_Week	INT64
Family_Income	STRING	Family_Income	STRING
Exam_Anxiety_Score	NUMERIC	Exam_Anxiety_Score	NUMERIC
Final_Score	STRING	Grade	STRING

Key Technical Operation: Tables were joined via a Student_ID key.

EXECUTIVE SUMMARY

The data reveals that academic success is a multi-variable equation where both psychological health and support systems are as predictive as students' natural academic aptitude. This project identifies 'Threshold Anxiety' at the C/D GPA border as a primary predictor of academic failure and utilized CTE-driven intervention analysis to evaluate institutional safeguards to propose targeted, data-driven support strategies.

- A moderate negative correlation ($r=-0.49$) exists between exam anxiety and final scores.
- Anxiety levels account for 24% of the variance in final grades ($r^2=0.24$), suggesting that nearly one-quarter of a student's performance is dictated by their stress state.
- Students who received formal support achieved an average final score of 77.32, representing a 20.87% increase over the non-tutored group average of 63.97.

INSIGHTS DEEP DIVE

The C/D Threshold Spike: Data reveals a significant non-linear spike in stress as students approach the passing threshold. Students in the "Aspiring D" category (those working to move from failing to a 1.7 GPA) reported anxiety levels 11.75% higher than those in the "Safe" category.

- **Insight:** The psychological pressure is highest at the point of "Academic Survival." Interestingly, anxiety levels actually decrease once students move into "At Risk A" or "Aspiring B" tiers, suggesting that once a basic level of security is reached, the intense "survival stress" begins to subside.
- While both "C/D Borderline" groups report high anxiety, the "Aspiring D" group shows significant emotional volatility (Sigma 1.99), suggesting a subset of students may be reaching a breaking point. Conversely, the "At Risk C" group shows a more uniform stress response (Sigma 1.46), identifying them as a prime candidate for group-based wellness interventions.

Tutoring as a Stabilizer: The data indicates that institutional intervention (tutoring) provides a dual benefit: it raises the academic "ceiling" while simultaneously stabilizing the anxiety levels of students.

- **Performance Lift:** Students who utilized tutoring achieved an average final score of 78.8, a 38.5% increase over the non-tutored average of 56.9.
- **Reduced Volatility:** The Score Sigma for tutored students was 10.48, compared to 14.77 for those without support.

- Conclusion: Tutoring acts as a "Predictability Engine." It results in a 31.86% reduction in performance variance, meaning tutored students not only score higher but are significantly less likely to experience "outlier" failing grades.

Lifestyle Safeguards

While sleep is generally a stabilizer, the data reveals a non-linear relationship where the "Poorly Rested" group achieved the highest average score (73.6). However, because this group consists of a single student (n=1), this represents a high-performing outlier rather than a scalable trend. It suggests a "crunch" effect where a specific individual prioritize short-term academic output over rest.

- The Performance Peak: The "Well Rested" group outperformed the "Moderately Rested" group by 14.3%. Notably, the "Well Rested" group exhibited the lowest Score Sigma (10.82), indicating that adequate sleep leads to the most consistent and predictable academic results. In contrast, the "Moderately Rested" group's performance was highly volatile (Sigma 20.73), showing a lack of stability in retention.
- The Anxiety Trade-off: The data shows a clear emotional tax associated with sleep levels. The "Well Rested" group maintained the lowest anxiety score (3.98). As sleep quality decreased to "Moderately Rested", anxiety scores spiked to 6.53—a 64.3% increase in stress. While the single "Poorly Rested" outlier maintained a moderately-high anxiety level (6.2), the broader trend suggests that "moderate" sleep deprivation is the primary catalyst for emotional dysregulation.
- The Moderately Rested Crisis: The most significant finding remains the "Moderately Rested" cohort. As the largest group (n=10), they represent the "worst of both worlds" scenario: they recorded the lowest average scores (60.81) paired with the highest average anxiety (6.53). This group likely represents students struggling with cognitive fog—getting enough sleep to be aware of their academic pressure, but not enough to effectively process or retain the information.
- Conclusion: For a rare subset of students, extreme sleep deprivation can coincide with high performance in the short term, though it remains statistically unreliable. For the majority, falling into Moderate Sleep Deprivation results in a measurable crisis: a significant decrease in scoring consistency and a massive spike in anxiety. The data suggests that Full Rest is the only reliable safeguard for maintaining both high academic performance and emotional stability.

RECOMMENDATIONS

- Allocate immediate counseling and coaching resources to students in the 1.6-1.8 GPA range (Aspiring D/At Risk C). Data indicates this cohort faces a 10.6%

(Aspiring D's spike in anxiety compared to "Safe" students, suggesting that the psychological pressure of the passing threshold is a primary barrier to their academic performance.

- Expand tutoring access. With tutored students showing a 20.87% higher average score and a 35.42% reduction in grade volatility, tutoring should be positioned as both an academic elevator and an emotional stabilizer that provides a consistent performance "floor".
- Institute measure to encourage sleep hygiene: Encourage better sleep habits for the Moderately Rested cohort to shift them toward Well Rested habits, a transition that correlates with a 14.3% score lift, a 39.1% reduction in anxiety, and nearly double the performance consistency.